# LAB 6: BRANCH COVERAGE ANALYSIS

**Gym Membership Application**

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Course: BTech – CSE (AI & ML)

Assigned Repository: L6_L7_21_gym_membership

GitHub Repository: https://github.com/Preeti1k3/L6_L7_21_gym_membership

Tools Used: Python 3.13.1, Pytest 8.3.4 and Coverage.py

## 1. Aim

To measure and improve branch coverage of the Gym Membership application using Coverage.py and Pytest by exercising the decision outcomes in the application.

## 2. Objective

The objective is to test the outcomes of conditional statements, including True and False decision paths and relevant boundary conditions, and achieve at least 90% branch coverage.

## 3. Application Under Test

The application file gym_membership.py contains membership_cost(), can_enter() and bmi_category(). These functions include decisions related to plan validity, age, discounts, trainer options, gym entry rules, height validation and BMI classification.

## 4. Branch Coverage

Branch coverage measures whether the possible outcomes of decision points in a program have been executed during testing. For example, for an if condition, tests should exercise both the condition being True and the condition being False where those outcomes exist.

## 5. Test Cases Added for Branch Coverage

A separate test file named BranchCoverageTest_21_gym_membership.py was created. The 30 test cases exercised the following conditions and decision outcomes:

- Valid and invalid membership plans.
- Age below 14, minimum valid age, senior age and below-senior age.
- Student and non-student conditions, including senior/student discount precedence.
- Trainer and non-trainer conditions for monthly, quarterly and yearly plans.
- Active and inactive membership conditions.
- Before-opening, opening-time boundary, closing-time boundary and after-closing cases.
- Female and male outcomes during women-only slots and entry during regular slots.
- Invalid zero and negative heights, and BMI values at category boundaries.

## 6. Branch Coverage Commands

```
coverage run --branch --source=gym_membership -m pytest -v
BranchCoverageTest_21_gym_membership.py
```

```
coverage report -m
```

## 7. Branch Coverage Results

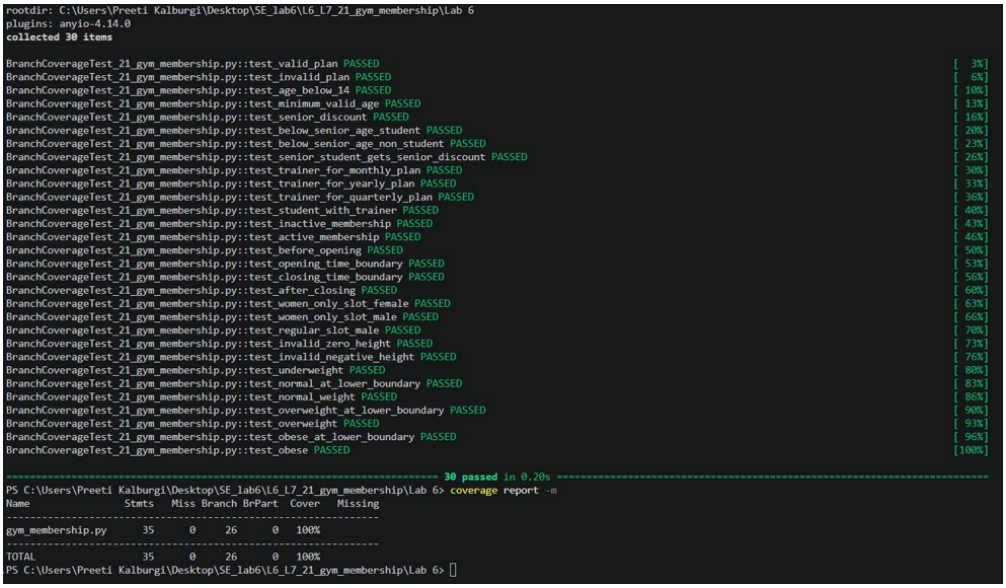
| Metric | Result |
|---|---|
| Tests passed | 30 |
| Total statements | 35 |
| Statements missed | 0 |
| Total branches | 26 |
| Partial branches | 0 |
| Branch coverage | 100% |

Analysis: All 30 tests passed. The coverage report showed 35 statements with no missed statements, 26 branches and zero partially covered branches. All reported branch outcomes were

exercised by the test suite, resulting in 100% branch coverage. This exceeds the required minimum of 90%.

## 8. Terminal Output Screenshot

Branch coverage test execution and coverage report:




## 9. Git Commit

Commit ID: 5dec398
Commit message: Add tests achieving 100 percent branch coverage
The branch coverage test file and its screenshot were committed and pushed to the personal GitHub fork.

## 10. Conclusion

Branch coverage analysis was performed on the Gym Membership application. The 30 additional test cases exercised all 26 reported branches, with zero partially covered branches. The final branch coverage achieved was 100%, satisfying and exceeding the lab requirement.